Parameters and architecture

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Parameters and architecture before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Forward passes by hand

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Activation functions

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Representing inputs as vectors

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Consider a dense network with 2 **input**s, 3 hidden units using ReLU, and 1 **output** unit using identity. We call its widths 2 to 3 to 1. These widths, connectivity and activation choices describe its **architecture**.

The weights and biases supply the numerical values used by that **architecture**.

A fully connected hidden unit receiving 2 **input**s needs 2 weights plus 1 bias. Three such units need 6 weights and 3 biases. The **output** receives 3 activations and needs 3 weights plus 1 bias. Total: 13 **parameter**s.

Input values and computed activations are not additional **parameter**s.

Layer	Weights	Biases	Total
2 inputs to 3 hidden units	2 times 3 = 6	3	9
3 hidden values to 1 output	3 times 1 = 3	1	4
Whole network	9	4	13

This counting rule assumes one separate bias per **output** unit and no shared or tied weights. Other **architecture**s can use different rules.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Parameters and architecture trace and record the first step you can explain without help.

Use W2: Sort the quantities for **Parameter**s and **architecture**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Parameters and architecture explanation against the supplied facts before discussing it.

Use the worked case for **Parameter**s and **architecture**. Proposed explanation: “Count one bias per **input**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Count a new architecture independently

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

If hidden width becomes 4, recount and explain why activation values are excluded.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Count a single layer

> Count a single layer. Apply the supplied Parameters and architecture facts and keep your reasoning.

A **dense layer** has 4 **input**s and 2 **output**s with one bias per **output**. Count weights, biases and total.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Sort the quantities

> Sort the quantities. Apply the supplied Parameters and architecture facts and keep your reasoning.

Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, **parameter**, **architecture** choice or **training** setting.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Change the bias convention

Change the bias convention. Apply the supplied Parameters and architecture facts and keep your reasoning.

Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Count a new architecture independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Count a new architecture independently. Apply the supplied Parameters and architecture facts and keep your reasoning.

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

If hidden width becomes 4, recount and explain why activation values are excluded.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Limit a size claim

> Limit a size claim. Apply the supplied Parameters and architecture facts and keep your reasoning.

A team says its 26-**parameter** model must outperform a 14-**parameter** model. Give a fair check and one reason the claim is unsupported.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Parameters and architecture to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled **parameter**s and **architecture evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish architectural choices from **parameter** values.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Parameters and architecture

> Count trainable weights and biases separately from activation functions. Explain the labelled step before checking the changed condition.

Count trainable weights and biases separately from activation functions.

Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/7

Trainable parameter count

	weights	biases	total
input to hidden	6	3	9
hidden to output	3	1	4

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/7

Parameters and architecture

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count first layer

2 x 3=6 weights plus 3 biases gives 9 parameters.



2. Count output layer

3 x 1=3 weights plus 1 bias gives 4 parameters.



3. Total

Total trainable parameters=9+4=13; ReLU adds no parameters here.

CHANGE ONE THING

Increase hidden width from 3 to4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Count trainable weights and biases separately from activation functions.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here.

1. Total

2. Count **output** layer

3. Count first layer

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Total Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here. Count one weight for each **input**-to-unit connection and one bias for each receiving unit.

Two **input**s connecting to three hidden units make six weights. The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Increase hidden width from 3 to4. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. First layer 12 **parameter**s; **output** layer 5; total 17.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: First layer 12 **parameter**s; **output** layer 5; total 17. Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Repair a missing building block

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network's final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

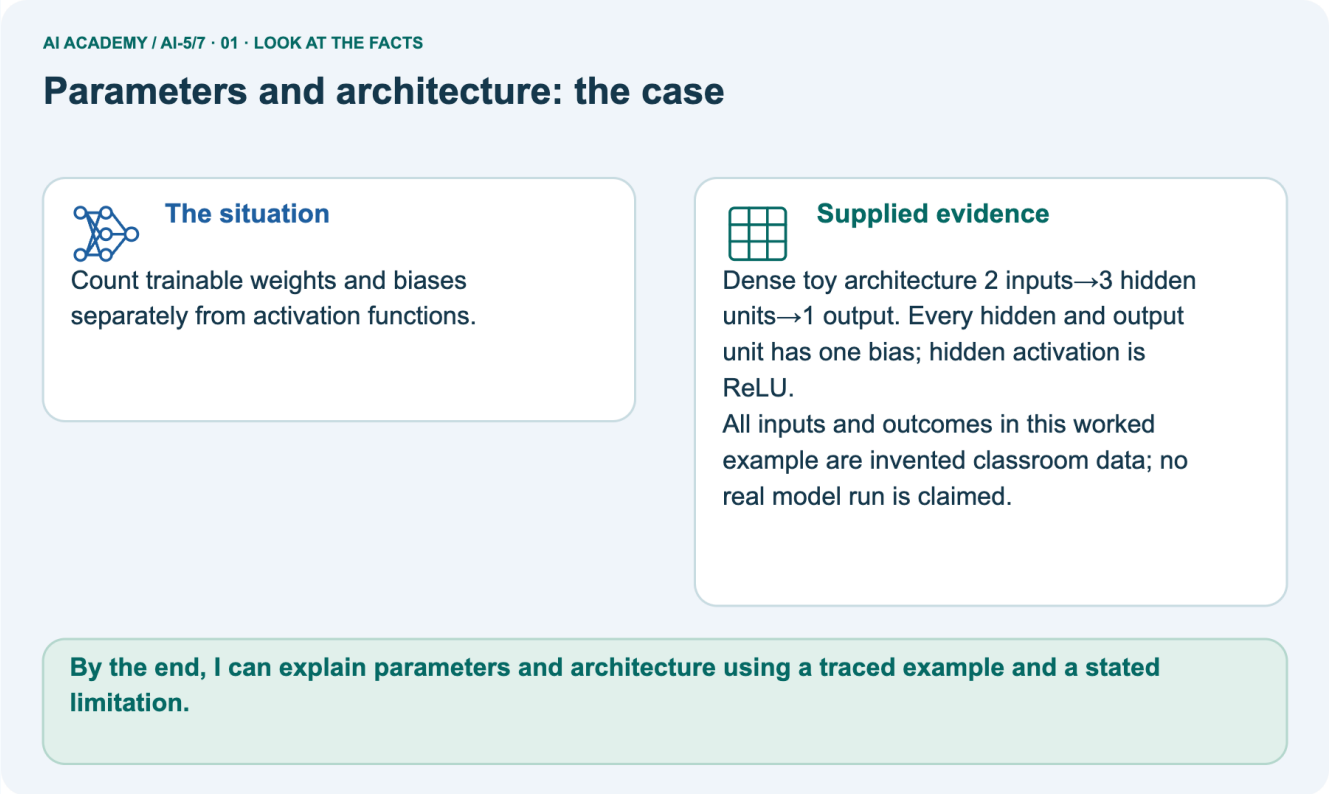
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Parameters and architecture picture from its labels, then record a relevant limit.

01. Parameters and architecture: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

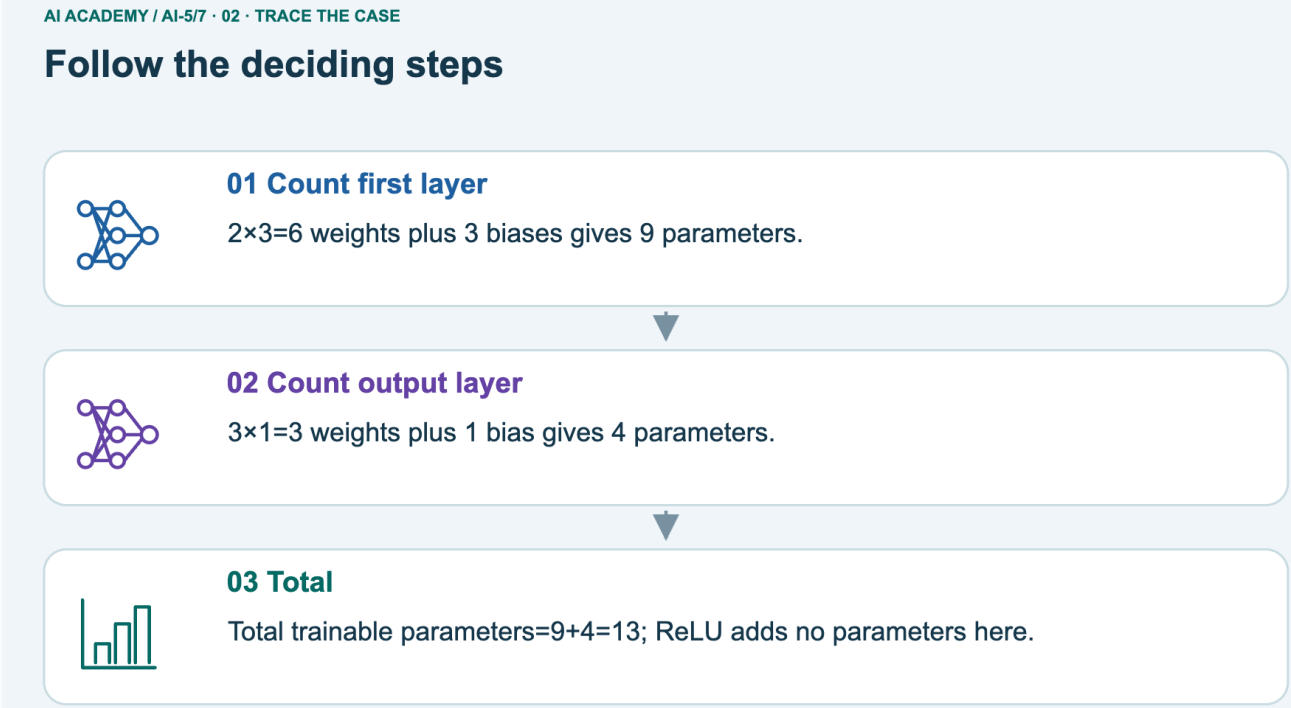
Count trainable weights and biases separately from activation functions. Supplied facts: Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



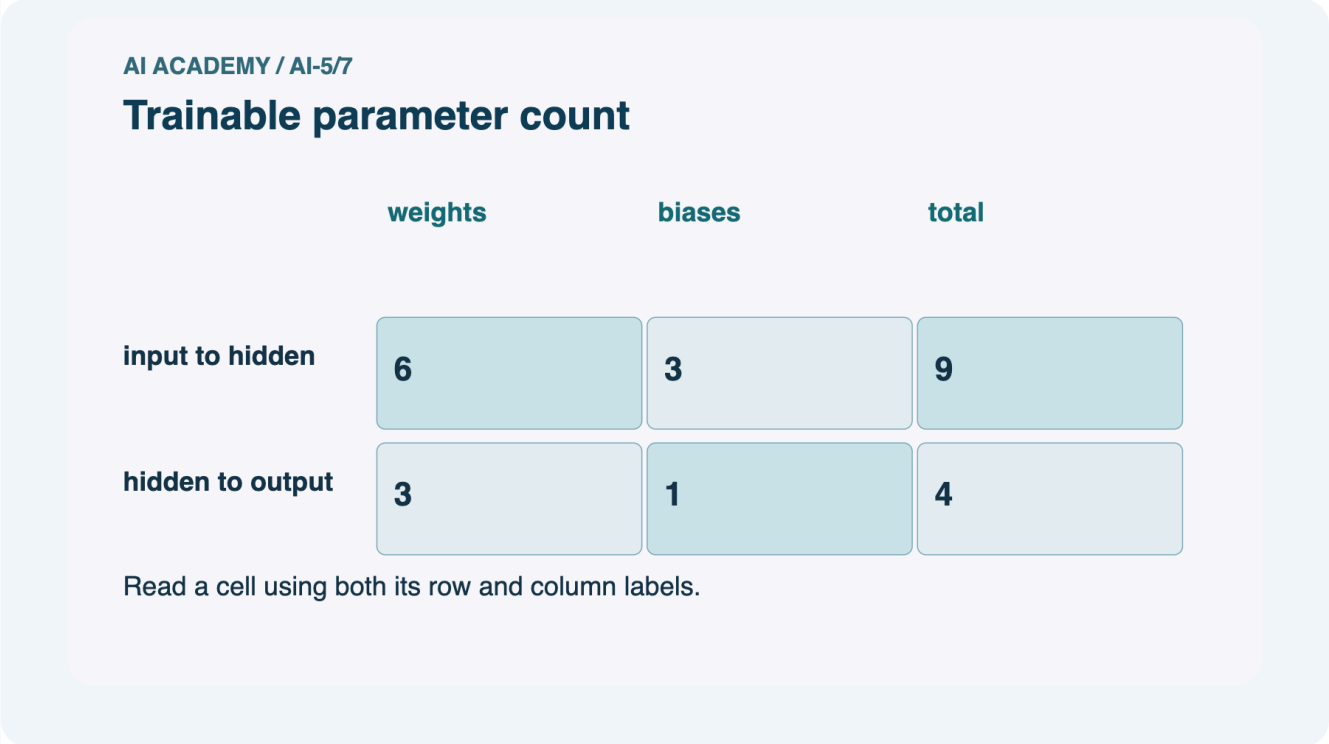
Follow the labelled picture with the matching explanation. Open the vector diagram

Count first layer: 2×3=6 weights plus 3 biases gives 9 **parameter**s. → Count **output** layer: 3×1=3 weights plus 1 bias gives 4 **parameter**s. → Total: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Trainable parameter count



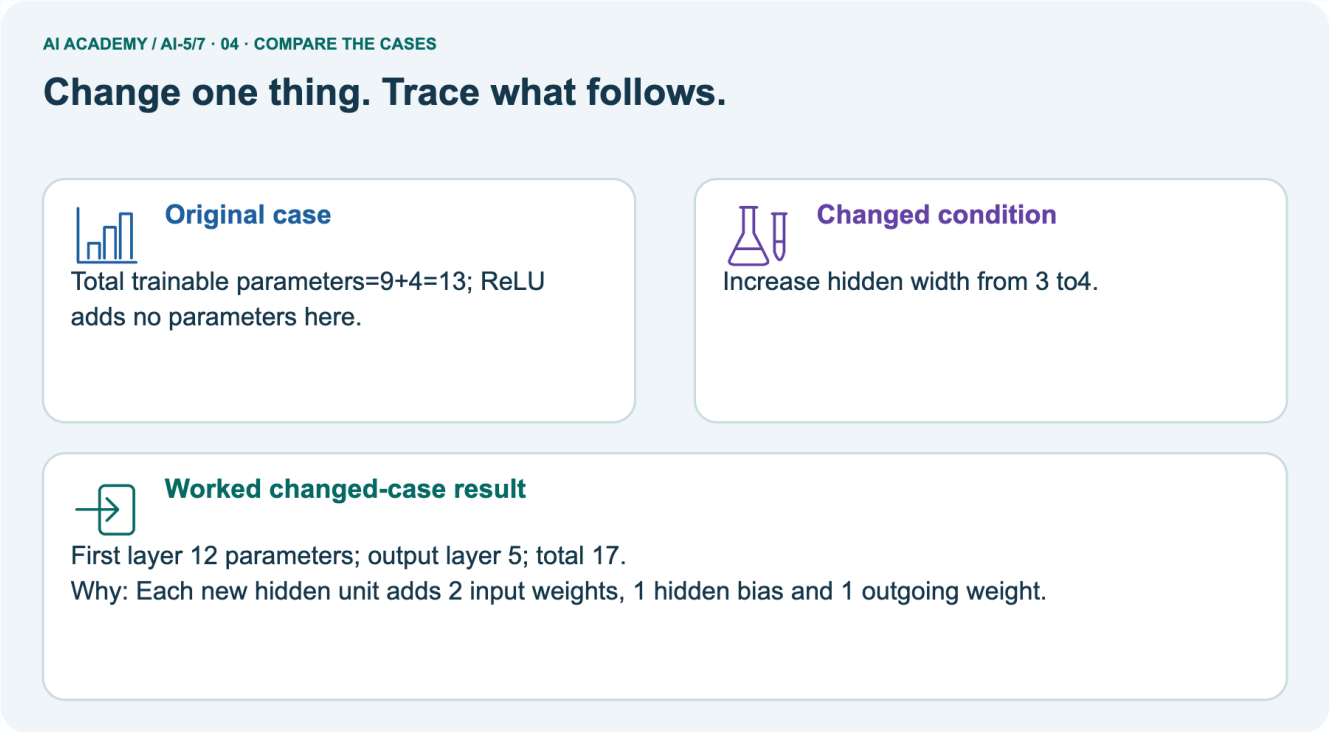
Follow the labelled picture with the matching explanation. Open the vector diagram

Trainable **parameter** count. **input** to hidden: weights = 6, biases = 3, total = 9. hidden to **output**: weights = 3, biases = 1, total = 4.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

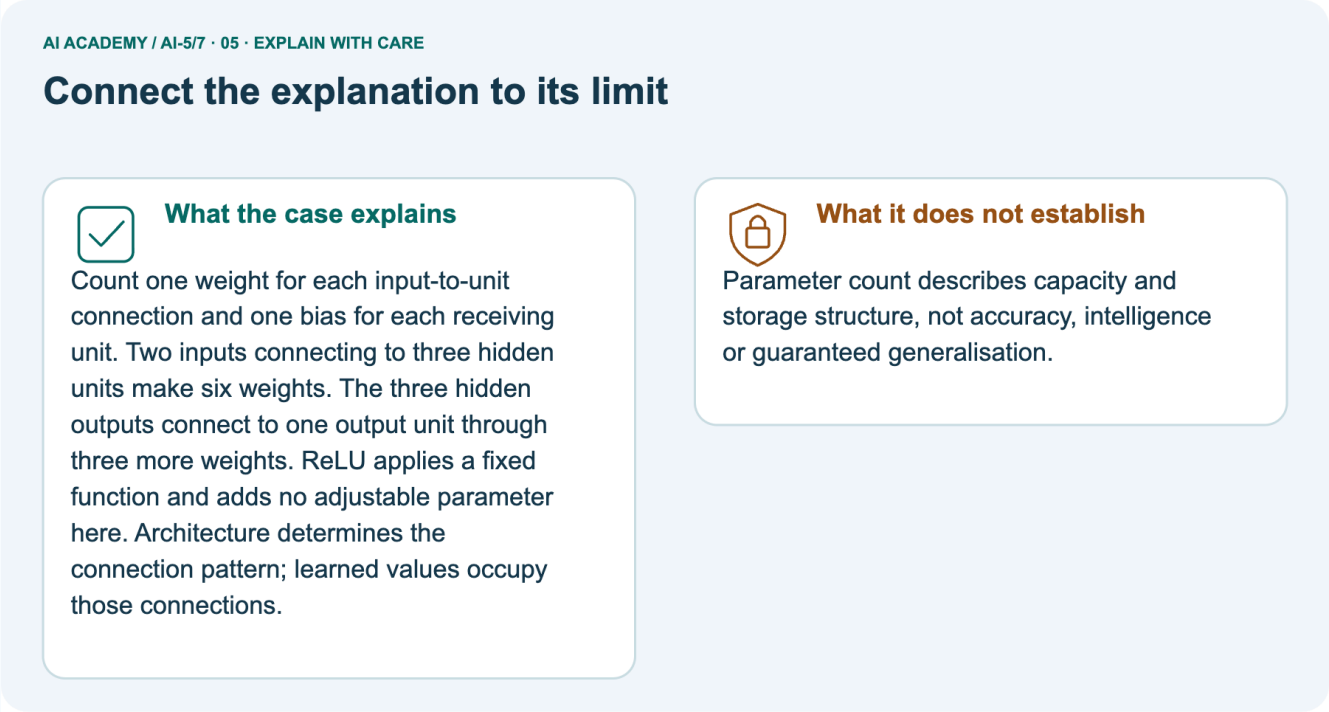
Original case: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here. Changed condition: Increase hidden width from 3 to4. Expected result: First layer 12 **parameter**s; **output** layer 5; total 17.

Reason: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Count one weight for each **input**-to-unit connection and one bias for each receiving unit. Two **input**s connecting to three hidden units make six weights.

The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections. Limit: **Parameter** count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

Explain the case, then point to a limit on the claim.

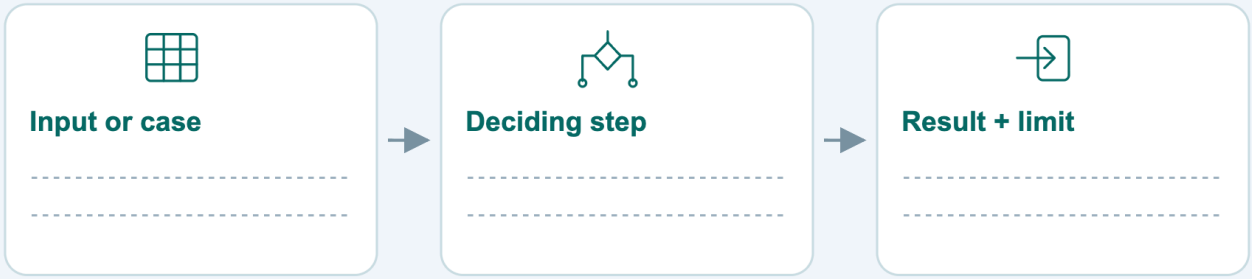
My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-5/7 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



How many new parameters did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.